

# Julius Alexander Zeiss, M.Sc. Physics

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## Profile

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- I develop classical algorithms that make difficult quantum problems tractable. Combining group theory, quantum Shannon theory, and convex optimization, I use quantum de Finetti theorems and symmetry-reduced semidefinite programming hierarchies to obtain rigorous bounds on quantum correlations and quantum error correction. Alongside these algorithmic questions, I study the power and limitations of entanglement in communication tasks and non-local games, especially quantum graph coloring. My work connects quantum information with graph theory and combinatorics, investigating how entanglement changes classical constraints and how graph structure shapes quantum strategies.
- Speaker at leading international conferences (TQC, Beyond IID, AQIS) and invited speaker at institutions including Stanford University and the Perimeter Institute for Theoretical Physics
- Host of a regular A.I. Workshop at RWTH Aachen on AI tools for rigorous, reproducible science

## Education

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| since 03/2023     | <b>PhD in Physics (Quantum Information Theory), RWTH Aachen University, Germany</b><br>Institute for Quantum Information, final-year PhD student<br>Advisor: Prof. Dr. Mario Berta                                                                                                                                        |
| 10/2019 – 08/2022 | <b>M.Sc. Physics, University of Cologne &amp; University of Bonn, Germany</b><br>Focus: Theoretical Particle Physics, General Relativity, Mathematical Physics<br>Thesis: <i>SDP Relaxations for Quantum Inflation</i><br>Advisor: Prof. Dr. David Gross, Thesis Grade: 1.0 (very good)<br>Overall Grade: 1.4 (very good) |
| 10/2018 – 08/2019 | <b>M.Sc. Physics, University of Münster, Germany</b><br>Focus: Theoretical Nuclear & Particle Physics, Theoretical Non-linear Physics, Quantum Information Theory, Biophysics                                                                                                                                             |
| 10/2015 – 09/2018 | <b>B.Sc. Physics, University of Münster, Germany</b><br>Thesis: <i>Numerical Treatment of the Manning Potential for the Description of Ammonia</i><br>Advisor: Prof. Dr. Gernot Münster, Thesis Grade: 1.0 (very good)<br>Overall Grade: 1.8 (good)                                                                       |
| 04/2013 – 09/2015 | <b>B.Sc. Business Administration, University of Münster, Germany</b><br>(discontinued, transferred to Physics)                                                                                                                                                                                                            |
| 09/2010 – 05/2012 | <b>International Baccalaureate, King William's College, Isle of Man</b><br>Overall Grade: 41/45 Points (very good, 1.1 German Abitur)                                                                                                                                                                                     |

## Publications and Preprints

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| 2026 | J. A. Zeiss, <i>A counterexample to the quantum Hedetniemi conjecture</i> , arXiv:2609.20690                                                                                              |
| 2026 | J. A. Zeiss, <i>Optimal entanglement-assisted source coding under a balanced-difference promise</i> , arXiv:2609.15757                                                                    |
| 2026 | M. Berta, P. Costa Rico, G. Koßmann, L. Lami, J. A. Zeiss, <i>Sharp continuity of quantum conditional entropy</i> , arXiv:2607.24687                                                      |
| 2026 | G. Koßmann, J. A. Zeiss, <i>Fixed points in de Finetti hierarchies</i> , arXiv:2607.23689 (under review at Communications in Mathematical Physics)                                        |
| 2026 | J. A. Zeiss, G. Koßmann, R. Schwonnek, M. Plávala, <i>Finite de Finetti for convex bodies and polynomial optimization</i> , arXiv:2601.15184 (under review at Mathematical Programming A) |
| 2025 | J. A. Zeiss, G. Koßmann, O. Fawzi, M. Berta, <i>Approximating fixed size quantum correlations in polynomial time</i> , arXiv:2507.12302 (under review at PRX Quantum)                     |
| 2025 | G. Koßmann, J. A. Zeiss, O. Fawzi, M. Berta, <i>On approximate quantum error correction for symmetric noise</i> , arXiv:2507.12326 (under review at Quantum)                              |

## Invited Talks

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| 06/2026 | <b>Stanford University, CS Theory Group, Palo Alto, USA</b><br><i>A Family of Information-theoretic de Finetti Theorems for Constrained Optimization</i>                                             |
| 04/2026 | <b>University of Cologne, Quantum Information Seminar, Germany</b><br><i>Approximating fixed size quantum correlations in polynomial time</i>                                                        |
| 07/2023 | <b>Siegen University, Germany</b><br><i>Inner approximation schemes for approximate quantum error correction</i>                                                                                     |
| 02/2023 | <b>Perimeter Institute for Theoretical Physics, Waterloo, Canada</b><br><i>Infinite Dimensional Optimisation Problems in Quantum Information — An operator algebra approach to the NPA Hierarchy</i> |
| 05/2022 | <b>Alpen-Adria-Universität Klagenfurt, Austria</b><br><i>SDP methods for the quantum causal compatibility problem</i>                                                                                |

## Contributed Talks

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| 09/2026 | <b>TQC 2026, Sherbrooke, Canada</b><br><i>A Family of Information-Theoretic de Finetti Theorems for Constrained Optimization</i>                                        |
| 05/2026 | <b>ILAS 2026, Virginia Tech, Blacksburg, USA</b><br><i>Convergent Inner-Outer Approximation Schemes From De Finetti Theorems For Games And Quantum Error Correction</i> |
| 03/2026 | <b>Quantum at the Dunes, IIP-UFRN, Natal, Brazil</b><br><i>Approximating fixed size quantum correlations in polynomial time</i>                                         |
| 08/2025 | <b>AQIS 2025, The University of Hong Kong, Hong Kong</b><br><i>Approximating fixed size quantum correlations in polynomial time</i>                                     |
| 07/2025 | <b>13th Beyond IID in Information Theory, TU Munich, Germany</b><br><i>Approximating fixed size quantum correlations in polynomial time</i>                             |

## Posters

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| 04/2026 | <b>851. WE-Heraeus-Seminar: 90 Years of Einstein-Podolsky-Rosen Paradox, Bad Honnef, Germany</b><br><i>Approximating fixed size quantum correlations in polynomial time</i>                              |
| 01/2026 | <b>QIP 2026, Riga, Latvia</b><br><i>Approximating fixed size quantum correlations in polynomial time</i>                                                                                                 |
| 08/2025 | <b>QMATH Masterclass on Representation Theory in Quantum Information Science, University of Copenhagen, Denmark</b><br><i>Schur–Weyl Duality for Efficient Approximation of Constrained Separability</i> |
| 07/2024 | <b>Beyond IID in Information Theory, University of Illinois Urbana-Champaign, USA</b><br><i>Approximating fixed size quantum correlations in polynomial time</i>                                         |

## Teaching

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| since 03/2023     | <b>RWTH Aachen University, Germany</b><br>Teaching and supervision of students at the Institute for Quantum Information     |
| 04/2021 – 04/2022 | <b>University of Cologne, Germany</b><br>Tutorials on Linear Algebra & Analysis, Proofs in Mathematics, Theoretical Physics |
| 04/2019 – 09/2019 | <b>University of Münster, Germany</b><br>Tutorials on Applied Physics                                                       |

## Academic Service

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| Conferences | TQC (2023, 2024, 2025) • ISIT (2023, 2024, 2025) • QIP (2024) • Beyond IID (2023, 2024) |
| Journals    | IEEE (2025)                                                                             |

## Outreach

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| since 2025  | <b>A.I. Workshop, RWTH Aachen University, Germany</b><br>Founder and host of a regular workshop on AI tools for research — from coding assistants and version control to formally verified proofs — and on using them to conduct better science |
| 2024 – 2026 | <b>ML4Q — Matter and Light for Quantum Computing, Germany</b><br>Instructor: school internships (July 2024), Quanteen Day 2024, Quanteen Day 2026                                                                                               |
| 11/2022     | <b>DPG Physik Journal</b><br>A. Matthies and J. Zeiss, <i>Quantum Computing</i> , review article on the Quantum Computing Bad Honnef Physics School, Physik Journal 21 (2022) No. 11, p. 57                                                     |

## Skills

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| Languages | German (native), English (bilingual)                          |
| Coding    | Python, Mathematica, Fortran, PHP, JavaScript, HTML, SQL, VBA |

## Outside of Academia — Employment

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|-------------------|--------------------------------------------------------------------------------------------------------|
| 2014, 2015        | <b>Fielmann AG, Hamburg, Germany</b> — DCF Modelling with Monte-Carlo Simulations, VBA & SQL Solutions |
| 01/2013 – 02/2013 | <b>HMT, Hamburg, Germany</b> — Mathematical Modelling using Monte-Carlo Simulations                    |
| 08/2011           | <b>AKO Capital, London, England</b> — Internship in Financial Research using Mathematical Modelling    |